Group C · Progression in year 3 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**49**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

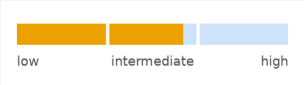
Sample: **TCGA-A2-A3XY-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.02** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

61st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

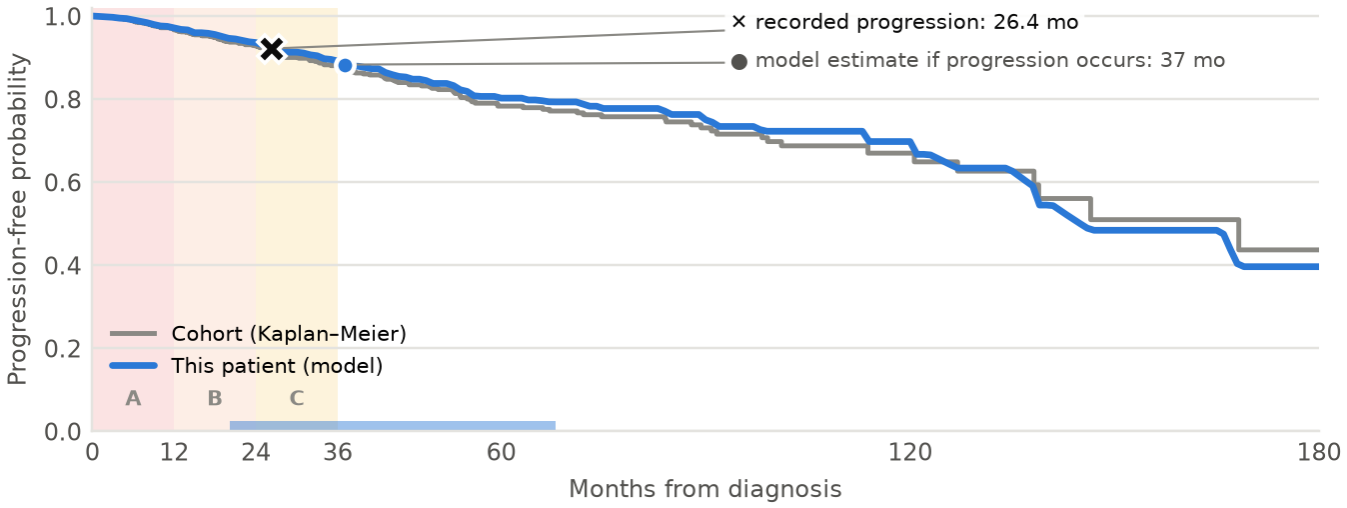
≈ 37 months

Window > 36 months · IQR 20–68 mo if progression occurs

Recorded progression

26.4 months

Group C · Progression in year 3
✗ recorded window differs · |error| 11 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 49-year-old patient has a subtype-unassigned tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **61st risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND 26.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **LSM1** to **mRNA decay by 5' to 3' exoribonuclease**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.66). Progression was recorded at 26.4 months; the model's estimate given progression was 37 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

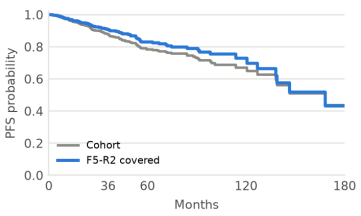


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A3XY** deleted → **LSM1** in pathway → **mRNA decay by 5' to 3' exoribonuclease**	84	↓ **slower progression** HR 0.66, z -1.3, n = 103 carriers of “mRNA decay by 5' to 3' exoribonuclease”	**0.67**
2 ★	**TCGA-A2-A3XY** deleted → **KCNU1** in pathway → **Sperm Motility And Taxes**	99	↓ **slower progression** HR 0.87, z -0.5, n = 97 carriers of “Sperm Motility And Taxes”	**0.66**
3 ★	**TCGA-A2-A3XY** deleted → **DDHD2** in pathway → **Synthesis of PA**	85	↓ **slower progression** HR 0.80, z -0.9, n = 189 carriers of “Synthesis of PA”	**0.64**
4 ★	**TCGA-A2-A3XY** deleted → **STAR** in pathway → **Pregnenolone biosynthesis**	87	↓ **slower progression** HR 0.84, z -0.7, n = 196 carriers of “Pregnenolone biosynthesis”	**0.62**
5 ★	**TCGA-A2-A3XY** amplified → **ARID4B** in pathway → **Potential therapeutics for SARS**	71	↑ **faster progression** HR 1.36, z +1.4, n = 145 carriers of “Potential therapeutics for SARS”	**0.61**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

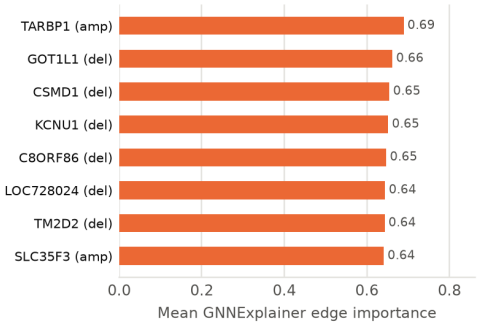
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FGFR1 · driver	Homozygous deletion	GISTIC ±2	43	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.31
TARBP1	High-level amplification	GISTIC ±2	0	0.69
GOT1L1	Homozygous deletion	GISTIC ±2	0	0.66
CSMD1	Homozygous deletion	GISTIC ±2	0	0.65
KCNU1	Homozygous deletion	GISTIC ±2	1	0.65
C8ORF86	Homozygous deletion	GISTIC ±2	0	0.65
LOC728024	Homozygous deletion	GISTIC ±2	0	0.64
TM2D2	Homozygous deletion	GISTIC ±2	0	0.64
SLC35F3	High-level amplification	GISTIC ±2	0	0.64
LETM2	Homozygous deletion	GISTIC ±2	0	0.63
PROSC	Homozygous deletion	GISTIC ±2	0	0.63



39 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 52 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1ES	35	II	missing	0.80	progressed at 113.8 mo (group D)
TCGA-E2-A1IE	61	II	LumA	0.75	progressed at 59.5 mo (group D)
TCGA-WT-AB41	55	II	LumA	0.74	no progression, followed 53.0 mo
TCGA-B6-A0WT	61	II	LumA	0.73	progressed at 127.0 mo (group D)
TCGA-BH-A204	80	II	LumB	0.73	no progression, followed 83.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.